LSTM and Bi-LSTM Based Deep Learning Approach to Detect Obsessive-Compulsive Disorder Using Multichannel EEG Signals

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***Abstract* - Obsessive-Compulsive Disorder (OCD) can be detected from multichannel electroencephalogram (EEG) signals using certain Deep Learning based frameworks such as Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (Bi-LSTM) networks. Although raw EEG data is very noisy and unstructured, the temporal dependencies across EEG signals from multiple brain regions are effectively captured by these models when a preprocessing pipeline which involves sampling, band-pass filtering (0-50 Hz), normalization, and segmentation is used to convert the raw data into structured multichannel time-series tensors. These tensors are suitable for model input, enabling the Binary Classification between OCD vs non-OCD subjects. This study demonstrates the real-world possibility of deep learning models, especially recurrent neural networks, in developing an automated and objective solution to OCD detection using EEG signals, achieving an accuracy of 91%, precision of 0.91, recall of 0.91 and F1-score of 0.91 on the test dataset using Bi-LSTM.**

I. INTRODUCTION:

Obsessive Compulsive Disorder (OCD) is a mental disorder that is marked by persistent thoughts (obsessions) and compulsive actions. It severely hampers and affects daily activities, affecting around 3.5% of people in the world and about 60% of people diagnosed with OCD experience chronic symptoms in their lives [4]. Diagnostic complexity and delays in clinical diagnosis are created as the OCD symptoms have an overlapping nature with other psychiatric disorders [1]. Electroencephalogram (EEG) is a neuro-imaging tool which uses sensors attached to the scalp to measure electrical activity inside the brain. It is highly accurate and gives precise results within milliseconds. Electroencephalograms provide data from several channels in the form of time-series data which can be measured in different frequency bands like delta, theta, alpha, beta and gamma waves. This non-invasive and low cost method has become a valuable tool for detecting neural signatures of psychiatric conditions [3][4].

With regard to OCD, EEG studies found anomalies in brain areas including frontotemporal cortex, as well as changes in frequency bands' activities. This variability suggests that EEG signals include discriminative features that could distinguish OCD patients from controls. Recent EEG analyses (e.g. in youth) report stronger resting state connectivity in frontal, temporal, and occipital networks of OCD patients especially in the gamma, beta and delta frequency bands [6][10]. Manual analysis of EEG data, however, poses significant challenges since it is high-dimensional, noisy, and non-linear, making the current methods for classification inefficient and subjective.

Earlier, the approaches were often dependent on highly engineered features and conventional Machine Learning models like Logistic SVMs which gave moderate results (≈85% accuracy) [2] and some recent EEG based ensembles can achieve about 90% accuracy [12]. In the recent literature, several researchers have applied machine learning/deep learning approaches on different kinds of data for detecting OCD patients. For example, Farhad et al. have demonstrated that hybrid deep learning based on EEG signals can be an effective approach to detect OCD with 1DCNN-LSTM providing 90.88% cross-validation accuracy and 98.11% test accuracy, and 1DCNN-GRU providing 85.91% cross-validation accuracy and 96.72% test accuracy.

In particular, recurrent neural networks like Long Short-Term Memory (LSTM) and bidirectional LSTM (Bi-LSTM) are well-suited for the sequential EEG data. Naderi and Jahanian-Najafabadi (2025) note that LSTM/Bi-LSTM architectures have become the most common RNN models in EEG based brain disorder studies (≈30% of DL approaches) because they capture temporal dependencies. Also since very large datasets of EEG are not readily available, models like LSTM prove to perform robustly with relatively smaller datasets. These models can thus preserve long-range temporal context that traditional methods might miss.

In this study, we implement LSTM and Bi-LSTM models for binary classification of OCD vs non-OCD subjects using multichannel EEG. The dataset contains 32 EEG channels with each channel consisting of 256 sequential samples in time-domain. The raw EEG is preprocessed through filtering, normalization, segmentation, and augmentation to form structured time-series inputs.

It should be noted that such a structure makes it easy to use both LSTM and BiLSTM models in the problem setting.The main task of the study is to implement two different deep learning models LSTM and BiLSTM for binary classification of OCD vs non-OCD subjects. Instead of using pre-engineered features for classification tasks, it is important to apply LSTM/Bi-LSTM models directly to raw EEG sequences.

In summary, it is a step towards improving computational psychiatry, particularly by using electroencephalography signals and using Deep Learning models. By tackling some problems in the area like data representation and model choice, this research seeks to develop a more scientific approach towards OCD diagnosis in the future proving that Deep Learning models like LSTM and Bi-LSTM are able to outperform the conventional Machine Learning models in analyzing such complex datasets.

II. LITERATURE REVIEW

REFERENCE	FOCUS OF STUDY	KEY METHODS	MAJOR FINDINGS
[1]	OCD vs Healthy classification	1DCNN-LSTM, 1DCNN GRU	1DCNN-LSTM achieved **90.88% CV accuracy**, external test **98.11%**; 1DCNN-GRU achieved **85.91% CV accuracy**, test **96.72%**
[2]	MRI/fMRI-based OCD diagnosis	SVM, Logistic Regression, Random Forest	SVM: 85%, LR: 70%, Random Forest: 65%.
[5]	OCD episode prediction using wearable biosensors .	Logistic Regression, RF, Neural Networks, MERF	RF and MERF achieved **70% accuracy** in real-world OCD event detection
[7]	EEG (resting)	2D-CNN on time–frequency EEG images vs. SVM	85.0% accuracy (AUC 0.88) with CNN; outperformed SVM (45.0% acc, AUC 0.47). CNN identified OCD vs. control.
[9]	OCD, SAD, ADHD detection	GaussianNB, RF, SVM	**91% accuracy**, precision **76%**, specificity **96%**
[13]	EEG (resting, children/adolescents)	CNN (VGG16 on EEG connectivity maps)	86.6% accuracy (F1 85.3%, specificity 94.6%, AUC 94.8%)

III. DATA AND METHODOLOGY

The main aim was to identify the OCD condition via EEG data acquired using multiple channels through the employment of deep learning algorithms, that is, LSTM and BiLSTM algorithms. The sequential properties of the EEG signal data set can be used by organizing each channel data into sequences of discrete time domain points, which are subsequently utilized as input features.

A. Dataset Overview:

The dataset contains 50000 EEG samples classified into 18 disorders. Out of these, 2765 were classified as OCD diagnosed. The dataset also gives the clinical info of people which includes age, sex, IQ and EQ of the samples.Further, it contains 32 columns of Electrode data, sampled at 256 Hz (Fig. 1) stored as a NumPy Array for every electrode. This means that 256 time steps are used for sampling the continuous time-series data, as it becomes the suitable format which can be used in LSTM and Bi-LSTM models. But the raw EEG data contains a lot of noise and needs to be filtered out first. Band pass filters (0-50 Hz) are used to remove extra noise from EEG signals and get the data relevant to us [1], [10]. Apart from this the signal needs to be scaled down and normalized to bring mean close to 0 and standard deviation close to 1.

B. Data Pre-Processing:

Data pre-processing was essential for the EEG electrode columns only. Brain signals collected were sampled at 256 Hz rate. In order to train the neural network effectively and to have stable convergence, the amplitude distribution of signals should be normalized [1], [13]. Firstly, the dataset had been saved in the CSV file where EEG signal segments were encoded in strings in arrays.

To make it possible to store and process the array data, Parquet was chosen to convert the CSV file into a more effective format that allows storing structured data types in one cell. Finally, the obtained string-encoded arrays were transformed into Python lists and further converted to NumPy arrays. Normalization of data was applied to prepare the EEG signals for the further use in the deep learning framework.

C. Target Labelling:

This dataset included 18 different classes of various neurological and psychological diseases such as attention deficit hyperactivity disorder, Parkinson's disease, bipolar disorder, Alzheimer's disease, and OCD. To achieve our objective, we used the binary classification approach to focus specifically on OCD detection. In particular, each OCD sample was given the label of 1 whereas all other diseases were labeled 0. This made it possible for the algorithm to detect differences between these two groups and classify more accurately.

D. Train-Test Split:

The dataset of 50k samples was split into training and testing data sets at an 80:20 split. To further test the performance of the models developed, a validation set was obtained through the partitioning of the training data, which involved using 20% of the training data. Hence, three sets of data were obtained from the data split for efficient learning.

E. Data Augmentation:

It was observed that the data contained a high level of imbalance in terms of the classes of OCD and non-OCD patients. In order to overcome this challenge, methods of data augmentation were employed on the data of the OCD category by employing methods such as amplification, adding Gaussian noise, and temporal shifting [1], [8]. The effect of this was an increase in the number of OCD data instances which balanced the class distribution, improving the overall accuracy, precision, recall and F1 score of the models due to the reduced overfitting of data.

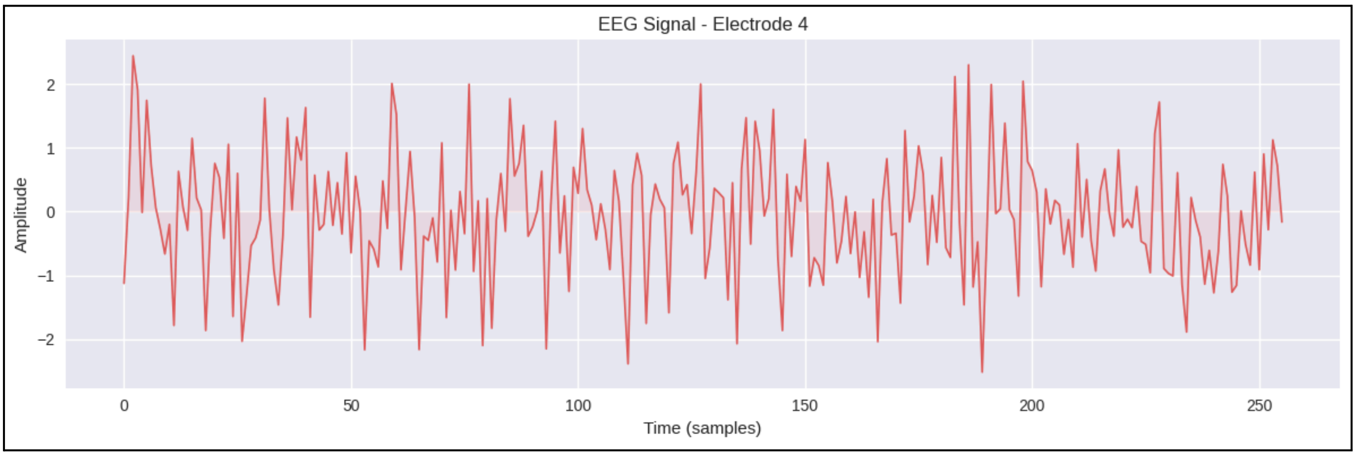
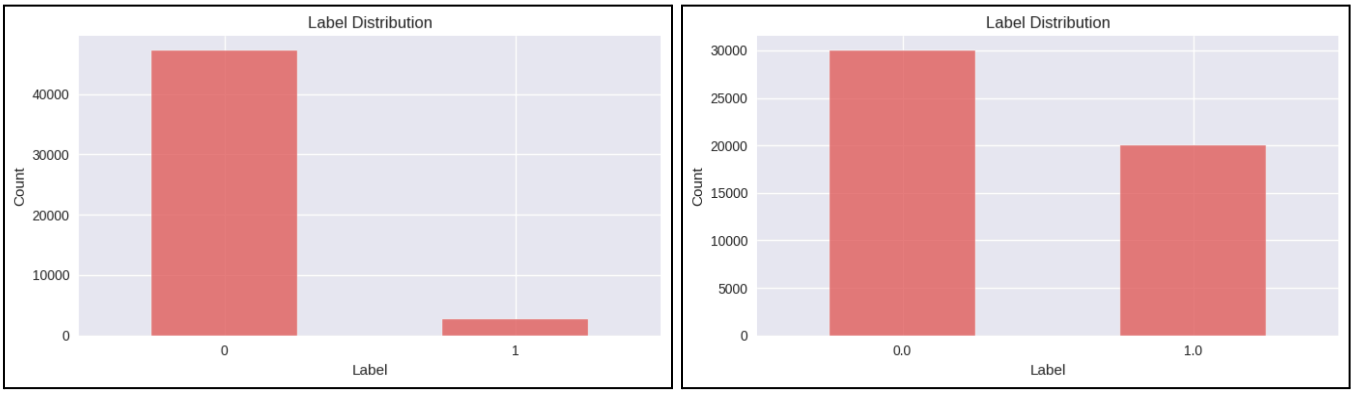


Fig. 1. Sampled EEG Amplitude signal from Electrode



Before Data Augmentation After Data Augmentation

Fig. 2. Comparison of Class distribution before and after data augmentation

IV. PROPOSED METHODOLOGY

The pipeline of the proposed method for classification based on the EEG signals starts with a raw dataset that contains about 50,000 rows including the EEG signals and the non-EEG related information. For making the raw dataset more diverse, data augmentation processes such as adding noise to the dataset, scaling, and temporal shifting of the EEG signals are used. Data augmentation processes will provide some variations similar to the actual world for improving the model generalization capabilities therefore increasing the overall count of the OCD samples. After performing data augmentation processes on the dataset, noise reduction, feature engineering and data cleaning operations are conducted to achieve better-quality data of signals. Moreover, non-EEG-related information is scaled in order to prepare for the next phase, and mainly to take the overview of the dataset.

At the feature transformation phase, the conversion of EEG data to a 3D tensor that takes time, frequency, and channel into consideration makes the data ready for deep learning-based algorithms. Then feature selection is carried out to select only important features while removing any redundant information. In the case of other metadata except EEGs, appropriate encoding mechanisms will be utilized to make the data compatible for machine learning. The split of the dataset into two subsets of training and testing datasets will be carried out with a split ratio of 80:20. The training and testing process is done for both machine learning and deep learning models for comparison. The study is based upon the LSTM and Bi-LSTM networks, showing how these RNN networks are capable of generating more accurate predictions in sequential data as compared to the conventional machine learning algorithms like Logistic Regression and Random Forest [1], [7], [13]. The final step will be the performance evaluation of these models using accuracy, precision, recall, F1-score, and AUC measures.

Figure 3 illustrates the entire process flow of the proposed method, giving a visual depiction of all the steps that take place in the EEG classification process. The flow diagram clearly shows the order in which the data is acquired, augmented, preprocessed, and transformed into features. It then proceeds to show how the model is trained and tested, and finally evaluated based on performance measure.

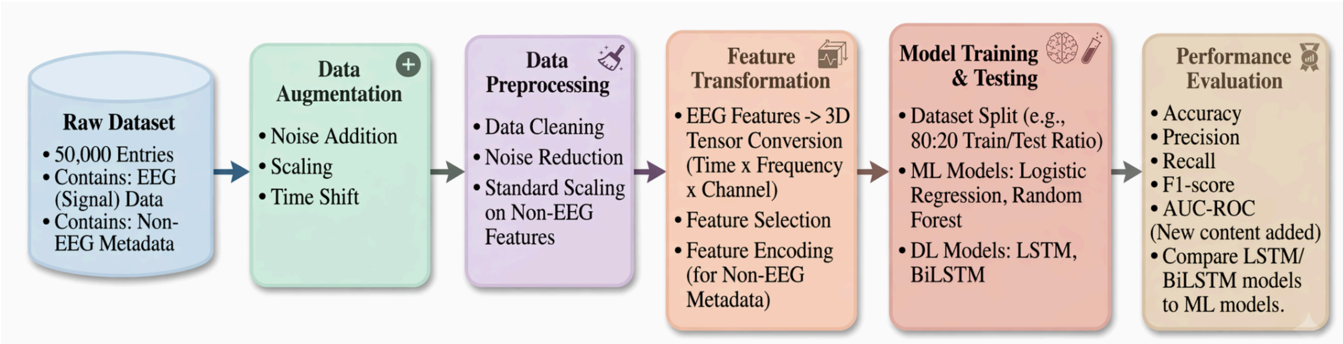


Fig. 3. Workflow Chart proposed

V. ML ALGORITHMS

Machine Learning algorithms enable systems to recognize patterns and make predictions without being explicitly programmed [6]. They are used in various use cases, like classification, regression, and pattern recognition; especially those areas where there is high dimensionality in the dataset, such as biomedicine.

A. Long Short-Term Memory Networks (LSTM)

Long Short-Term Memory is a recurrent neural network architecture designed specifically to learn from sequential data by capturing long-term dependencies [1], [16]. While typical RNNs tend to have difficulties with the problem of vanishing and exploding gradients [16], LSTMs solve this issue by using a gating mechanism that allows them to remember or disregard certain information.

A recurrent LSTM network is composed of multiple memory cells, where each of them has a cell state representing the long-term memory of the LSTM cell. The gates control the amount of information flowing into and out of the cell. The forget gate decides whether or not to retain the information in the previous cell state or disregard it. Similarly, the input gate manages what kind of information will be included in the current cell state, whereas the output gate specifies what information flows into the next hidden state. LSTM networks preserve only useful information for a long period of time, making them an excellent choice when working with time-series data [1], [7], [13]. For instance, this study uses LSTMs to extract temporal features of multi-channel EEG signals.

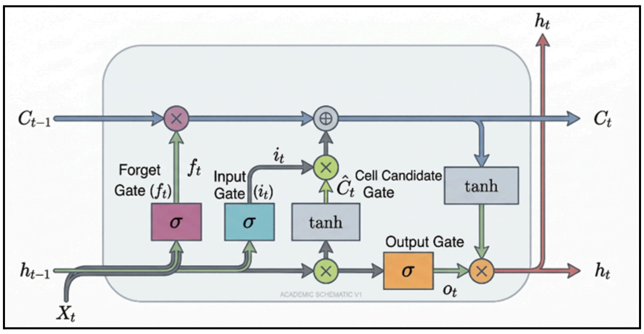


Fig. 4. LSTM Networks

1) Input Gate:

The input gate in an LSTM network controls how much new information is added to the cell state. It takes the current input and the previous hidden state to determine the importance of incoming data. A sigmoid activation function generates a gating vector with values between 0 and 1, deciding what information to retain or discard.

At the same time, a candidate cell state is created using a tanh function, representing new information that could be stored [16].

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (1)$$

Where x_t denotes the input at time step t, h_{t-1} is the previous hidden state, W_i and b_i represent the weight matrix and bias of the input gate, and σ is the sigmoid activation function.

2) Cell State Update:

The cell state update represents how the LSTM combines past memory with new information at each time step. The previous cell state is first scaled by the forget gate, which determines what portion of past information should be preserved. Simultaneously, the new candidate information regulated by the input gate is added to the cell state. This combination allows the network to maintain long-term dependencies while continuously updating its memory with relevant features [16].

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (2)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (3)$$

Where $\tilde{C}_t$ is the candidate cell state, and C_t represents the cell state update equation.

3) Forget Gate:

The forget gate in an LSTM network determines which information from the previous cell state should be retained or discarded. It takes the current input and the previous hidden state as inputs and passes them through a sigmoid activation function to generate a gating vector with values between 0 and 1.

These values act as filters, where values close to 1 retain important information and values close to 0 remove irrelevant or outdated data from the cell state [16]. This mechanism allows the network to forget unnecessary information and focus on relevant long-term dependencies.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (4)$$

4) Output Gate:

In LSTM, the output gate determines the amount of cell state that should be exposed as the hidden state in the current time. Rather than passing all information, it selectively filters only those things needed to pass at the present time and in relation to previous hidden states through the use of a sigmoid function. This results in a gating vector whose values lie between 0 and 1.

The cell state is then passed through a **tanh activation function** to scale its values between −1 and 1. The output gate selectively filters this transformed cell state through element-wise multiplication, producing the final hidden state [16].

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (5)$$

$$h_t = o_t \cdot tanh(C_t) \quad (6)$$

Where, o_t represents the Output power, h_t is the hidden state, W_o represents the Weight matrix of output gate and b_o is the bias of the output gate.

B. Bidirectional Long Short-Term Memory Networks (Bi-LSTM)

The Bi-LSTM is an expansion of the regular LSTM, where not only does it process the sequence and information from both the past and the future time steps [14]. The architecture of the BiLSTM consists of two separate LSTM layers. The first one processes the sequence in a regular manner, i.e., from t = 1 to t = T, outputting the forward hidden states h_t^f, here f denotes the forward side. In parallel, the second layer processes the same sequence but in reverse order (backward), i.e. from t = T to t = 1, generating backward hidden states h_t^b.The final hidden representation at each time step is obtained by combining the outputs from both directions, typically through concatenation:

$$h_t = [h_t^f; h_t^b] \quad (7)$$

This combined representation allows the network to utilize both preceding and succeeding context, which is particularly useful in tasks where the interpretation of a data point depends on its surrounding sequence [1], [13], [12].

Each directional LSTM independently maintains its own cell state and gating mechanisms, following the standard LSTM formulations described earlier. However, their parameters are not shared, allowing each direction to learn distinct temporal patterns and predict the values based on both forward and backward data learned.

$$h_t^f = LSTM_f(x_t, h_{t-1}^f, C_{t-1}^f) \quad (8)$$

$$h_t^b = LSTM_f(x_t, h_{t+1}^b, C_{t+1}^b) \quad (9)$$

The final output at each time step is then derived from the combined hidden state:

$$y_t = f(h_t) \quad (10)$$

where, f(·) is typically a fully connected layer followed by an activation function depending on the task

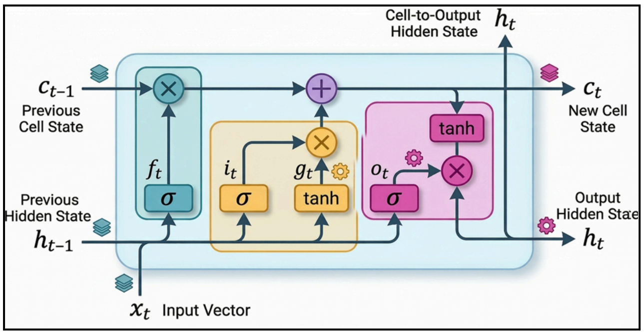


Fig. 5. Bi-LSTM Networks

VI. EXPERIMENTS AND RESULTS

Performance evaluation metrics are essential for evaluating the efficiency of classification algorithms and comparing different models to verify our results. These metrics are obtained using the confusion matrix, which represents prediction results of the model in all the cases i.e. the predictions are classified into four types: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), giving us the information about correct predictions and error types. The confusion matrix collectively helps in understanding both the capabilities and limitations of the classification model.

We evaluated the model on the widely used performance metrics such as Accuracy, Precision, Recall, F-1 Score, and Area Under the Curve (AUC), the models' performance is calculated to evaluate the model’s ability to correctly classify the cases labelled as 0 and 1. These measures are extensively used to ensure a fair and standardized comparison between different classification models as they give an unbiased evaluation of the model's classification performance in each of the confusion matrix's cases. Mathematical expressions for each metric are given below in Table I:

TABLE I:
FORMULAS FOR THE PERFORMANCE METRICS

S. No.	Metric	Formula
1.	Accuracy Rate	$\frac{TP+TN}{TP+TN+FP+FN}$
2	Precision	$\frac{TP}{TP+FP}$
3.	Recall	$\frac{TP}{TP+FN}$
4.	F1 Score	$\frac{2\times Precision\times Recall}{Precision+Recall}$

Evaluating the models on metrics shown in Table-I, training on a variety of models gives highly variant results as shown in Table-II. The LSTM and Bi-LSTM actually gave good results for sequential data and outperformed the traditional Machine Learning Models. The results highlight the effects of these deep learning models in capturing temporal dependencies in sequential EEG signal data.

For comparison, the experiment was also performed on the Random Forest classification algorithm, which performed well, giving an accuracy of 66%, and also on Logistic Regression, which recorded 50%. This difference is due to the ensemble nature of Random Forest, combining multiple decision trees trained on different subsets of the dataset, reducing variance and improving reliability. The Logistic Regression did not perform well in terms of precision(0.52), recall(0.50) and F1-Score(0.50). Random forest performed decently in terms of precision(0.76), recall(0.66) and F1-Score(0.58). The AUC for Logistic Regression and the Random Forest was 0.49 and 0.63 respectively which indicates poor classification. Logistic Regression, being a simple linear classifier, shows poor performance across all measures mentioned above, suggesting that linear classifiers may not be suitable for modeling EEG signals. This further highlights the limitation of traditional approaches in handling noisy physiological and high-dimensional data such as EEG signals.

The RNN models outperformed traditional machine learning approaches. The LSTM model achieved a good accuracy of 88%, while Bi-LSTM improved it even further to accuracy of 91%. This is due to the bidirectional nature of Bi-LSTM, enabling it to capture temporal dependencies in both past and future [1], [8]. LSTM achieved precision, recall, and F1-score of 0.88, while Bi-LSTM achieved 0.91. Both models showed balanced performance across both labels (0 and 1). This balanced performance indicates that the models are not biased toward any particular class and maintain a good consistency across predictions. Furthermore, the AUC score is 0.94 for LSTM and 0.96 for Bi-LSTM, indicating strong and clear class separation. Overall, these results show that deep learning networks, particularly Bi-LSTM, are highly effective for EEG-based OCD classification, while traditional models remain limited in handling complex temporal patterns. These findings also suggest the potential applicability of such models in real-world clinical monitoring systems to detect other disorders as well.

TABLE II:
COMPARISON OF LSTM AND BI-LSTM WITH CONVENTIONAL MACHINE LEARNING MODELS

Metric	**Logistic Regression**	**Random Forest**	**LSTM**	**Bi-LSTM**
Accuracy (%)	50.0	66.0	88.0	91.0
Precision	0.52	0.76	0.88	0.91
Recall	0.50	0.66	0.88	0.91
F1-Score	0.50	0.58	0.88	0.91
AUC	0.49	0.63	0.94	0.96

VII. CONCLUSION

Conclusively, this research has shown a good example of how useful EEG measurements combined with other features related to demographics could be in recognizing Classifying OCD vs non-OCD patients. It is clear from the current findings how significant it is to carry out proper preprocessing of collected data and adequate representation in order to get useful patterns out of it. It has become evident that the current approach is highly generalized and relatively reliable. This study has proven the validity of using Deep Learning models to learn sequential EEG data as means of diagnosing OCD and analysing its characteristics. This means that it might minimize the importance of subjectivity in diagnosis processes.

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